# FOP ALL PROGRAMS

## Assignment 1

### Program:

```c
#include<stdio.h>
void main()
{
int year;
printf("Enter year: ");
scanf("%d",&year);
if ((year%4==0 && year%100!=0) ||(year%400==0))
{
printf("%d is a leap year. \n", year);
}
else
{
printf("%d is not a leap year. \n", year);
}
}

```

### Output:

**Output**

```
Enter year: 1990
1990 is not a leap year.
```

**Output**

```
Enter year: 2024
2024 is a leap year.
```

# Assignment 2

## Program:

```c
#include <stdio.h>
void main()
{
int a, b, choice;
float r;
printf("SIMPLE CALCULATOR: \n 1:Addition \n 2:Subtraction \n 3
    :Multiplication \n 4:Division \n");
printf("Enter Choice: ");
scanf("%d", &choice);
printf("Enter two numbers: ");
scanf("%d %d", &a, &b);
switch(choice){
case 1:
{
r=a+b;
printf("%f", r);
}
break;
case 2:
```

```c
{
r=a-b;
printf("%f", r);
}
break;
case 3:
{
r=a*b;
printf("%f", r);
}
break;
case 4:
{
if (b==0)
{
printf("Divisor cannot be 0");
}
else
{
```

```
r=(float)a/b;
printf("%f", r);
}
}
break;
default:
printf("Error:Invalid Choice");
}
}
```

Output:

```
SIMPLE CALCULATOR:
 1:Addition
 2:Subtraction
 3:Multiplication
 4:Division
Enter Choice: 1
Enter two numbers: 234 563
797.000000
```

```
SIMPLE CALCULATOR:
 1:Addition
 2:Subtraction
 3:Multiplication
 4:Division
Enter Choice: 3
Enter two numbers: 124 42
5208.000000
```

# Assignment 3

## Program:

```c
#include<stdio.h>
void main()
{
int n, r, i=1;
printf("Enter number: ");
scanf("%d", &n);
while (i<=10)
{
r=n*i;
printf("%d \n", r);
i++;
}
}

```

## Output:

**Output**

```
Enter number: 67
67
134
201
268
335
402
469
536
603
670
```

# Assignment 4

## Program:

```c
#include <stdio.h>

int main() {
    int bp;
    float hra, ta, gs, pt, ns;
    printf("Enter Basic Pay: ");
    scanf("%d",&bp);
    hra=0.1*bp;
    ta=0.05*bp;
    gs=bp+hra+ta;
    printf("HRA component: %f\n", hra);
    printf("TA component: %f\n", ta);
    printf("Gross Salary: %f\n", gs);
    pt=0.02*gs;
    ns=gs-pt;
    printf("Net Salary:%f", ns);

    return 0;
}
```

## Output:

**Output**

```
Enter Basic Pay: 60000
HRA component: 6000.000000
TA component: 3000.000000
Gross Salary: 69000.000000
Net Salary:67620.000000

=== Code Execution Successful ===
```

# Assignment 5

## Program:

```c
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], sum[10][10];
    int i, j, k, r, c, choice;

    printf("Enter rows and columns: ");
    scanf("%d%d", &r, &c);

    printf("Enter Matrix A:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&a[i][j]);

    printf("Enter Matrix B:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&b[i][j]);
```

```c

    while(1)
    {
        printf("\n1.Addition\n2.Saddle Point (A)\n3.Inverse (A
            - 2x2)\n4.Magic Square (A)\n5.Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                for(i=0;i<r;i++)
                {
                    for(j=0;j<c;j++)
                    {
                        sum[i][j] = a[i][j] + b[i][j];
                        printf("%d ", sum[i][j]);
                    }
                    printf("\n");
```

```
            }
            break;

        case 2:
        {
            int found = 0;

            for(i=0;i<r;i++)
            {
                int min = a[i][0], col = 0;

                for(j=1;j<c;j++)
                {
                    if(a[i][j] < min)
                    {
                        min = a[i][j];
                        col = j;
                    }
                }
```

```

                for(k=0;k<r;k++)
                {
                    if(a[k][col] > min)
                        break;
                }

                if(k==r)
                {
                    printf("Saddle point = %d\n", min);
                    found = 1;
                }
            }

            if(!found)
                printf("No saddle point\n");

            break;
        }
```

```c

        case 3:
        {
            if(r!=2 || c!=2)
            {
                printf("Inverse only possible for 2x2 matrix\n");
                break;
            }

            float det;
            det = a[0][0]*a[1][1] - a[0][1]*a[1][0];

            if(det == 0)
            {
                printf("Inverse not possible\n");
            }
            else
            {
```

```c
                printf("Inverse matrix:\n");
                printf("%.2f %.2f\n", a[1][1]/det, -a[0][1]/det);
                printf("%.2f %.2f\n", -a[1][0]/det, a[0][0]/det);
            }
            break;
        }

        case 4:
        {
            if(r != c)
            {
                printf("Not a square matrix\n");
                break;
            }

            int sum = 0, flag = 1;

```

```
            for(j=0;j<c;j++)
                sum += a[0][j];

            for(i=0;i<r;i++)
            {
                int row=0, col=0;

                for(j=0;j<c;j++)
                {
                    row += a[i][j];
                    col += a[j][i];
                }

                if(row != sum || col != sum)
                    flag = 0;
            }

            int d1=0, d2=0;
            for(i=0;i<r;i++)
```

```
            {
                d1 += a[i][i];
                d2 += a[i][r-i-1];
            }

            if(d1 != sum || d2 != sum)
                flag = 0;

            if(flag)
                printf("Magic square\n");
            else
                printf("Not a magic square\n");

            break;
        }

        case 5:
            return 0;

```

```
        default:
            printf("Invalid choice\n");
    }
}
}

```

Output:

**Output**

```
Enter rows and columns: 2 2
Enter Matrix A:
3 5
6 2
Enter Matrix B:
4 7
1 9

1.Addition
2.Saddle Point (A)
3.Inverse (A - 2x2)
4.Magic Square (A)
5.Exit
Enter choice: 1
7 12
7 11
```

# Assignment 6A

## Program:

```c
#include <stdio.h>
long int factorial(int n)
{
    if(n == 0 || n == 1)
        return 1;
    else
        return n * factorial(n - 1);
}
int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    printf("Factorial = %ld\n", factorial(num));

    return 0;
}
```

## Output:

**Output**

```
Enter a number: 25
Factorial = 7034535277573963776
```

**Output**

```
Enter a number: 5
Factorial = 120
```

# Assignment 6B

## Program:

```c
#include <stdio.h>
long int factorial(int n)
{
    long int fact = 1;
    int i;
   for(i = 1; i <= n; i++)
        fact = fact * i;
    return fact;
}
int main()
{
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);
    printf("Factorial = %ld\n", factorial(num));
    return 0;
}

```

## Output:

**Output**

```
Enter a number: 25
Factorial = 7034535277573963776
```

**Output**

```
Enter a number: 5
Factorial = 120
```

# Assignment 7

## Program:

```c
#include <stdio.h>

struct student
{
    char name[50];
    int roll;
    float m1, m2, m3;
    float total, per;
};

int main()
{
    struct student s[10];
    int i, n;

    printf("Enter number of students: ");
    scanf("%d", &n);

```

```c
    for(i = 0; i < n; i++)
    {
        printf("\nEnter details of student %d\n", i+1);

        printf("Name: ");
        scanf(" %[^\n]", s[i].name);

        printf("Roll No: ");
        scanf("%d", &s[i].roll);

        printf("Marks in 3 subjects: ");
        scanf("%f %f %f", &s[i].m1, &s[i].m2, &s[i].m3);

        s[i].total = s[i].m1 + s[i].m2 + s[i].m3;
        s[i].per = s[i].total / 3;
    }

    printf("\n Student Results \n");

```

```c
    for(i = 0; i < n; i++)
    {
        printf("\nName: %s", s[i].name);
        printf("\nRoll No: %d", s[i].roll);
        printf("\nTotal: %.2f", s[i].total);
        printf("\nPercentage: %.2f\n", s[i].per);
    }

    return 0;
}

```

## Output:

**Output**

```
Enter number of students: 2

Enter details of student 1
Name: Anand
Roll No: 2
Marks in 3 subjects: 45 65 87

Enter details of student 2
Name: Nalini
Roll No: 5
Marks in 3 subjects: 76 69 82
```

```
 Student Results

Name: Anand
Roll No: 2
Total: 197.00
Percentage: 65.67

Name: Nalini
Roll No: 5
Total: 227.00
Percentage: 75.67
```

# Assignment 8

## Program:

```c
#include <stdio.h>

int main() {
    int m1, m2, m3, m4, m5, T;
    float P;

    printf("Enter 5 marks: ");
    scanf("%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5);

    if (m1 < 40 || m2 < 40 || m3 < 40 || m4 < 40 || m5 < 40) {
        printf("FAIL\n");
    } else {
        T = m1 + m2 + m3 + m4 + m5;
        P = T / 5.0;

        printf("Total: %d, Percentage: %.2f%%\n", T, P);
        printf("Result: PASS - ");

        if (P >= 75) {
            printf("Distinction\n");
        } else if (P >= 60) { // No need to check P < 75, the 'else' already implies it
            printf("First Division\n");
        } else if (P >= 50) {
            printf("Second Division\n");
        } else {
            printf("Third Division\n");
        }
    }

    return 0;
}
```

## Output:

**Output**

```
Enter 5 marks: 59 82 66 72 90
Total: 369, Percentage: 73.80%
Result: PASS - First Division
```

# Assignment 9

## Program:

```c
#include <stdio.h>
int main() {
    int a, b, c, d=0;
    printf("Enter number: ");
    scanf("%d", &a);
    b=a;
    while(b>0)
    {
    c=b%10;
    d=d+(c*c*c);
    b=b/10;
    }
    if (a==d){
    printf("Armstrong Number");
    }
    else{
    printf("Not Armstrong Number");
    }
    return 0;}
```

## Output:

**Output**

```
Enter number: 20
Not Armstrong Number
```

**Output**

```
Enter number: 153
Armstrong Number
```

# Assignment 10

## Program:

```c
#include <stdio.h>
#include <math.h>
int main()
{
int a, b, n, choice, i;
unsigned long long f = 1;
float r;
printf("SIMPLE CALCULATOR: \n 1:Addition \n 2:Subtraction \n 3
    :Multiplication \n 4:Division \n 5:Power \n 6:Factorial \n"
    );
printf("Enter Choice: ");
scanf("%d", &choice);
if (choice >= 1 && choice <= 5)
{
printf("Enter two numbers: ");
scanf("%d %d", &a, &b);
}
else if (choice==6)
{
```

```c
printf("Enter number: ");
scanf("%lld", &f);
}
switch(choice){
case 1:
{
r=a+b;
printf("%f", r);
}
break;
case 2:
{
r=a-b;
printf("%f", r);
}
break;
case 3:
{
r=a*b;
```

```
printf("%f", r);
}
break;
case 4:
{
if (b==0)
{
printf("Divisor cannot be 0");
}
else
{
r=(float)a/b;
printf("%f", r);
}
}
break;
case 5:
{
r=pow(a,b);
```

```
}
break;
case 6:
{
if (n<0)
{
printf("Error");
}
else
{
for (i = 1; i <= n; ++i) {
f *= i;
}
printf("Factorial of %d = %llu", n, f);
}
}
default:
{
printf("Error:Invalid Choice");
```

```
}
return 0;
}

```

Output:

**Output**

```
SIMPLE CALCULATOR:
 1:Addition
 2:Subtraction
 3:Multiplication
 4:Division
 5:Power
 6:Factorial
Enter Choice: 1
Enter two numbers: 45 78
123.000000
```

**Output**

```
SIMPLE CALCULATOR:
 1:Addition
 2:Subtraction
 3:Multiplication
 4:Division
 5:Power
 6:Factorial
Enter Choice: 4
Enter two numbers: 56 8
7.000000
```

# Assignment 11

## Program:

```c
#include <stdio.h>
#include <math.h>

int main() {
    int num, i;
    long long fact = 1;

    printf("Enter a number: ");
    scanf("%d", &num);


    if (num >= 0)
        printf("Square root = %.2f\n", sqrt(num));
    else
        printf("Square root not defined for negative numbers\n"
            );


    printf("Square = %d\n", num * num);
```

```c


     printf("Cube = %d\n", num * num * num);


     if (num <= 1) {
         printf("The number is NOT PRIME\n");
     } else {
         int isPrime = 1;
         for (i = 2; i <= num / 2; i++) {
             if (num % i == 0) {
                 isPrime = 0;
                 break;
             }
         }
         if (isPrime)
             printf("The number is PRIME\n");
         else
             printf("The number is NOT PRIME\n");
```

```c
    }


    if (num >= 0) {
        for (i = 1; i <= num; i++) {
            fact *= i;
        }
        printf("Factorial = %lld\n", fact);
    } else {
        printf("Factorial not defined for negative numbers\n");
    }

    if (num > 1) {
        printf("Prime factors: ");


        while (num % 2 == 0) {
            printf("2 ");
            num /= 2;
```

```c
        }


        for (i = 3; i <= sqrt(num); i += 2) {
            while (num % i == 0) {
                printf("%d ", i);
                num /= i;
            }
        }

        if (num > 2)
            printf("%d", num);

        printf("\n");
    } else {
        printf("No prime factors\n");
    }
    return 0;
}
```

Output:

**Output**

```
Enter a number: 34
Square root = 5.83
Square = 1156
Cube = 39304
The number is NOT PRIME
Factorial = 4926277576697053184
Prime factors: 2 17
```

**Output**

```
Enter a number: 456
Square root = 21.35
Square = 207936
Cube = 94818816
The number is NOT PRIME
Factorial = 0
Prime factors: 2 2 2 3 19
```

# Assignment 12

## Program:

```c
#include <stdio.h>

int smallestDivisor(int n) {
    if (n <= 1)
        return n;

    for (int i = 2; i <= n; i++) {
        if (n % i == 0)
            return i;
    }
    return n;
}

int gcd(int a, int b) {
    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }
    return a;
}

int main() {
    int num1, num2;

    printf("Enter two numbers: ");
    scanf("%d %d", &num1, &num2);

    int sd1 = smallestDivisor(num1);
    int sd2 = smallestDivisor(num2);
    int result_gcd = gcd(num1, num2);

    printf("Smallest divisor of %d = %d\n", num1, sd1);
    printf("Smallest divisor of %d = %d\n", num2, sd2);
    printf("GCD of %d and %d = %d\n", num1, num2, result_gcd);

    return 0;
}
```

Output:

**Output**

```
Enter two numbers: 3 6
Smallest divisor of 3 = 3
Smallest divisor of 6 = 2
GCD of 3 and 6 = 3
```

**Output**

```
Enter two numbers: 56 92
Smallest divisor of 56 = 2
Smallest divisor of 92 = 2
GCD of 56 and 92 = 4
```

# Assignment 13

## Program:

```c
#include <stdio.h>

int main() {
    int num, remainder;

    printf("Enter a number: ");
    scanf("%d", &num);

    printf("Digits in reverse order: ");

    while (num != 0) {
        remainder = num % 10;
        printf("%d ", remainder);
        num = num / 10;
    }

    return 0;
}
```

## Output:

**Output**

```
Enter a number: 64634
Digits in reverse order: 4 3 6 4 6
```

**Output**

```
Enter a number: 91352
Digits in reverse order: 2 5 3 1 9
```

# Assignment 14

## Program:

```c
#include <stdio.h>
#include <math.h>

int main() {
    int binary, decimal = 0, base = 1, remainder;
    printf("Enter a binary number: ");
    scanf("%d", &binary);

    while (binary > 0) {
        remainder = binary % 10;
        decimal += remainder * base;
        binary = binary / 10;
        base = base * 2;
    }

    printf("Decimal equivalent = %d\n", decimal);
    return 0;
}
```

## Output:

**Output**

```
Enter a binary number: 11
Decimal equivalent = 3
```

**Output**

```
Enter a binary number: 342
Decimal equivalent = 22
```

# Assignment 15

## Program:

```c
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main() {
    int i;

    srand(time(0));

    printf("Pseudo-random numbers:\n");

    for (i = 0; i < 5; i++) {
        int num = rand();
        printf("%d\n", num);
    }

    return 0;
}
```

## Output:

**Output**

```
Pseudo-random numbers:
1098185392
51498684
1980790149
1849889750
495645347
```

**Output**

```
Pseudo-random numbers:
1015184307
757544137
2125169059
1860248339
180077889
```

# Assignment 16

## Program:

```c
#include <stdio.h>

int main() {
    int n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n], even[n], odd[n];
    int e = 0, o = 0;

    printf("Enter %d integers:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);

        if (arr[i] % 2 == 0) {
            even[e++] = arr[i];
        } else {
            odd[o++] = arr[i];
```

```c
        }
    }

    printf("Even numbers: ");
    for (i = 0; i < e; i++) {
        printf("%d ", even[i]);
    }

    printf("\nOdd numbers: ");
    for (i = 0; i < o; i++) {
        printf("%d ", odd[i]);
    }

    return 0;
}
```

Output:

**Output**

```
Enter number of elements: 2
Enter 2 integers:
34
87
Even numbers: 34
Odd numbers: 87
```

**Output**

```
Enter number of elements: 4
Enter 4 integers:
234
12
21
67
Even numbers: 234 12
Odd numbers: 21 67
```

# Assignment 17

## Program:

```c
#include <stdio.h>

int main() {
    int n;
    float x, term, sum = 0;

    printf("Enter value of x (in radians): ");
    scanf("%f", &x);

    printf("Enter number of terms: ");
    scanf("%d", &n);

    term = x;
    sum = term;

    for (int i = 1; i < n; i++) {
        term = -term * x * x / ((2 * i) * (2 * i + 1));
        sum += term;
    }

    printf("Sum of sine series = %f\n", sum);

    return 0;
}
```

## Output:

**Output**

```
Enter value of x (in radians): 30
Enter number of terms: 3
Sum of sine series = 198030.000000
```

# Assignment 18

## Program:

```c
#include <stdio.h>
#include <string.h>

int main() {
    char str[100], str2[100], rev[100];
    int i, len;

    printf("Enter a string: ");
    scanf("%s", str);

    // Length
    len = strlen(str);
    printf("Length = %d\n", len);

    // Reverse
    for(i = 0; i < len; i++) {
        rev[i] = str[len - i - 1];
    }
    rev[len] = '\0';
```

```c
    printf("Reversed string = %s\n", rev);

    // Palindrome
    if(strcmp(str, rev) == 0)
        printf("Palindrome\n");
    else
        printf("Not Palindrome\n");

    // Equality check
    printf("Enter another string: ");
    scanf("%s", str2);

    if(strcmp(str, str2) == 0)
        printf("Strings are equal\n");
    else
        printf("Strings are not equal\n");

    // Substring check
    if(strstr(str, str2))
```

```
        printf("Substring found\n");
    else
        printf("Substring not found\n");

    return 0;
}
```

Output:

**Output**

```
Enter a string: Amazing
Length = 7
Reversed string = gnizamA
Not Palindrome
Enter another string: Kanak
Strings are not equal
Substring not found
```

**Output**

```
Enter a string: kanak
Length = 5
Reversed string = kanak
Palindrome
Enter another string: kango
Strings are not equal
Substring not found
```

# Assignment 19

## Program:

```c
#include <stdio.h>
#include <string.h>
struct Employee {
    char name[50];
    char designation[50];
    char gender;
    char doj[20];
    float salary;
};
int main() {
    struct Employee e[100];
    int n, i, male = 0, female = 0, highSalary = 0, asstManager
        = 0;
    printf("Enter number of employees: ");
    scanf("%d", &n);
    for(i = 0; i <= n; i++) {
        printf("\nEnter details of employee %d\n", i+1);
        scanf("%s %s %c %s %f",
              e[i].name, e[i].designation,
```

```c
              &e[i].gender, e[i].doj, &e[i].salary);
        if(e[i].gender == 'M')
            male++;
        else if(e[i].gender == 'F')
            female++;
        if(e[i].salary > 10000)
            highSalary++;
        if(strcmp(e[i].designation, "AsstManager") == 0)
            asstManager++;
    }
    printf("\nTotal employees = %d", n);
    printf("\nMale = %d, Female = %d", male, female);
    printf("\nSalary > 10000 = %d", highSalary);
    printf("\nAsst Manager = %d", asstManager);
    return 0;
}
```

Output:

**Output**

```
Enter number of employees: 2

Enter details of employee 1
Rakesh VP M 12-Apr-20 50000

Enter details of employee 2
Sneha AsstManager 23-Jan-22 35000

Total employees = 2
Male = 1, Female = 0
Salary > 10000 = 2
Asst Manager = 1
```

**Output**

```
Enter number of employees: 2

Enter details of employee 1
Sagar AVP M 20-Feb-12 15000

Enter details of employee 2
Neela AsstManager F 11-Mar-16 30000

Total employees = 2
Male = 1, Female = 1
Salary > 10000 = 2
Asst Manager = 1
```

# Assignment 20A

## Program:

```c
#include <stdio.h>
void swap(int a, int b)
{
    printf("before swapping: a = %d, b = %d\n", a, b);
    int temp;
    temp = a;
    a = b;
    b = temp;
    printf("after swapping: a = %d, b = %d\n", a, b);
}
int main()
{
    int x, y;
    printf("Enter two numbers: ");
    scanf("%d %d", &x, &y);
    swap(x, y);
    printf("after swapping: x = %d, y = %d\n", x, y);
    return 0;
}
```

## Output:

**Output**

```
Enter two numbers: 6 9
before swapping: a = 6, b = 9
after swapping: a = 9, b = 6
after swapping: x = 6, y = 9
```

# Assignment 20B

## Program:

```c
#include <stdio.h>

void swap(int *a, int *b)
{
    int temp;
    temp = *a;
    *a = *b;
    *b = temp;
}

int main()
{
    int x, y;
    printf("Enter two numbers: ");
    scanf("%d %d", &x, &y);
    swap(&x, &y);
    printf("After swapping: x = %d, y = %d\n", x, y);
    return 0;
}
```

## Output:

**Output**

```
Enter two numbers: 6 9
After swapping: x = 9, y = 6
```

# Assignment 21

## Program:

```c
#include <stdio.h>
int main() {
    FILE *fp;
    char ch;
    fp = fopen("fp.txt", "w");
    fprintf(fp, "Hello World\n");
    fprintf(fp, "Welcome to Pune\n");
    fclose(fp);

    fp = fopen("fp.txt", "r");
    printf("File contents:\n");

    while ((ch = fgetc(fp)) != EOF) {
        putchar(ch);
    }

    fclose(fp);
    return 0;
}
```

## Output:

```
File contents:
Hello World
Welcome to Pune
```

# Assignment 22

## Program:

```
built in str
#include <stdio.h>
#include <string.h>

int main()
{
    char str1[100], str2[100], temp[100];
    int choice, i, len;
    printf("Enter first string: ");
    scanf("%s", str1);
    printf("Enter second string: ");
    scanf("%s", str2);

    while(1)
    {
        printf("\n STRING OPERATIONS MENU \n");
        printf("1. Find Length (String 1)\n");
        printf("2. Copy Strings \n");
        printf("3. Concatenate Strings\n");
```

```
        printf("4. Compare Strings\n");
        printf("5. Reverse String 1\n");
        printf("6. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                printf("Length of String 1 = %lu\n", strlen(str1
                    ));
                break;

            case 2:
                strcpy(temp, str1);
                strcpy(str1, str2);
                strcpy(str2, temp);
                printf("After copying:\n");
```

```c
            printf("String 1 = %s\n", str1);
            printf("String 2 = %s\n", str2);
            break;

        case 3:
            strcpy(temp, str1);
            strcat(temp, str2);
            printf("Concatenated string = %s\n", temp);
            break;

        case 4:
            if(strcmp(str1, str2) == 0)
                printf("Strings are equal\n");
            else
                printf("Strings are not equal\n");
            break;

        case 5:
            strcpy(temp, str1);
```

```c
            len = strlen(temp);

            for(i = 0; i < len / 2; i++)
            {
                char ch = temp[i];
                temp[i] = temp[len - i - 1];
                temp[len - i - 1] = ch;
            }
            printf("Reversed string = %s\n", temp);
            break;

        case 6:
            return 0;

        default:
            printf("Invalid choice!\n");
    }
}
}
```

Output:

**Output**

```
Enter first string: Shocking
Enter second string: Undecided

 STRING OPERATIONS MENU
1. Find Length (String 1)
2. Copy Strings
3. Concatenate Strings
4. Compare Strings
5. Reverse String 1
6. Exit
Enter your choice: 2
After copying:
String 1 = Undecided
String 2 = Shocking
```

```
 STRING OPERATIONS MENU
1. Find Length (String 1)
2. Copy Strings
3. Concatenate Strings
4. Compare Strings
5. Reverse String 1
6. Exit
Enter your choice: 1
Length of String 1 = 9

 STRING OPERATIONS MENU
1. Find Length (String 1)
2. Copy Strings
3. Concatenate Strings
4. Compare Strings
5. Reverse String 1
6. Exit
Enter your choice: 3
Concatenated string = UndecidedShocking
```